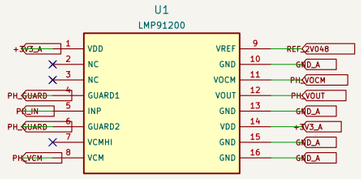
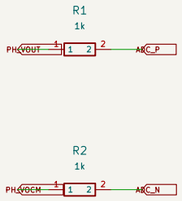
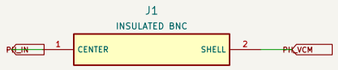
SOIL / pH AFE Rev A | Bench engineering prototype

5V regulated input / 3.3V I2C / external ESP32-S3 host / no onboard galvanic isolation

01 GUARDED ELECTRODE BUFFER









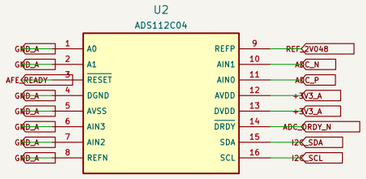


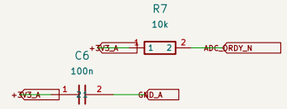
BNC shell = driven PH_VCM. Insulate from chassis. Floating sample only.

PH_IN: no test point, TVS or ordinary probe. Exposed guards on both faces; clean and dry.



02 DIFFERENTIAL CONVERSION



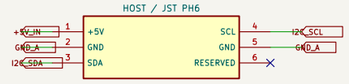




ADC: 0x40 / AIN0-AIN1 / gain2 / ext 2.048V / 20SPS

Register image: 02 02 00 00. Wait >=500ms after power; check readback.

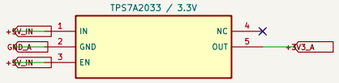
03 HOST INTERFACE

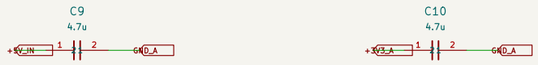




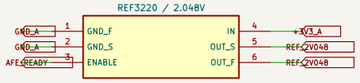


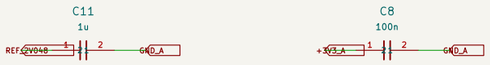
04 QUIET 3.3V SUPPLY



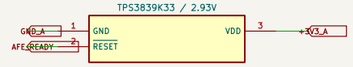


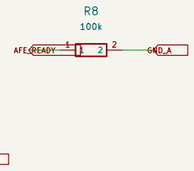
05 KELVIN REFERENCE



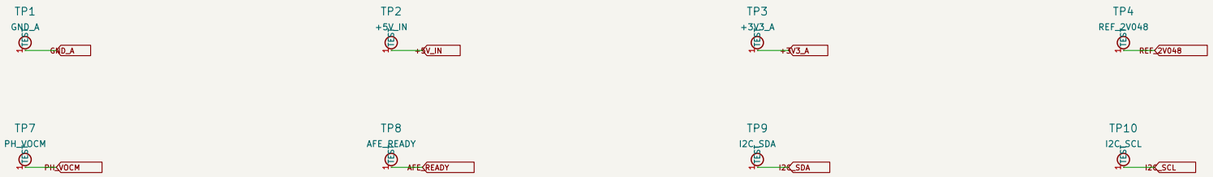


06 DELAYED ENABLE / RESET





07 LOW-IMPEDANCE TEST ACCESS | Ground clip: TP1 | No common-mode filter caps | R5/R6 removable pullup links



Engineering prototype — hardware validation pending

Sheet: /
File: Soil_pH_AFE.kicad_sch

Title: Soil pH Acquisition Module

Size: A3	Date: 2026-10-20	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/1